

Figure 1: Main features of AngularJS

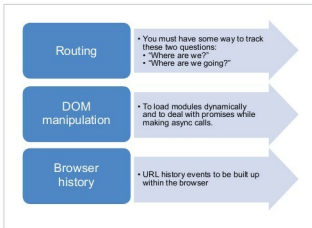


Figure 2: Challenges with SPAs

ng-app: This attribute stands for the active portion of the entire page. As per our code, it is the whole document. If you want to activate some specific location, then apply the ng-app attribute to that specific location tag.

ng-model and two-way data binding: ng-model binds the model value with the state of the text box. As per our code, we have defined a model as *yourname*. This value is attached with the value of the textbox using the ng-model attribute. So the change in the text box value will automatically reflect in the model *yourname*. This is nothing but two-way data binding. Similarly, if the model value is changed, then Angular will change the value of the text box.

The output of the 'Hello World' example is given in Figure 4.

Basic directives

Directives are nothing but the markers on a DOM element, which tell the compiler to attach a specified event or behaviour to that element.

Angular comes with a set of built-in directives. Some of these are described below.

Data Binding	- Two way Data binding Supported - Automatic synchronization of data between model and view components
Routing	- Built-in Routing Support - Allows switch between the views
Templates	Rendered view with information
Dependency Injection	Making components reusable, maintainable and testable
Directives	Allows to create new custom widgets
Deep Linking	Allows you to encode the state of application in the URL so that it can be bookmarked
Filters	Subset of items from an array and returns a new array

Figure 3: Core features of AngularJS



Figure 4: Output of 'Hello World'

- ng-app:** Initiates an AngularJS application
- ng-init:** Initialises application data
- ng-model:** Defines the model to be used in AngularJS
- ng-repeat:** Repeats HTML elements for each item in a collection
- ng-bind:** Binds the data to the element and is an alternative to the interpolation directive
- ng-show:** Shows the HTML element
- ng-hide:** Hides the model element
- ng-switch:** Used to add or remove elements from DOM, based on data
- ng-if:** Same as switch, just has a simpler syntax
- ng-include:** Includes the HTML elements from other files

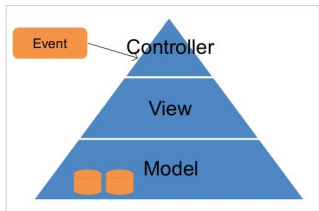


Figure 5: MVC architecture